

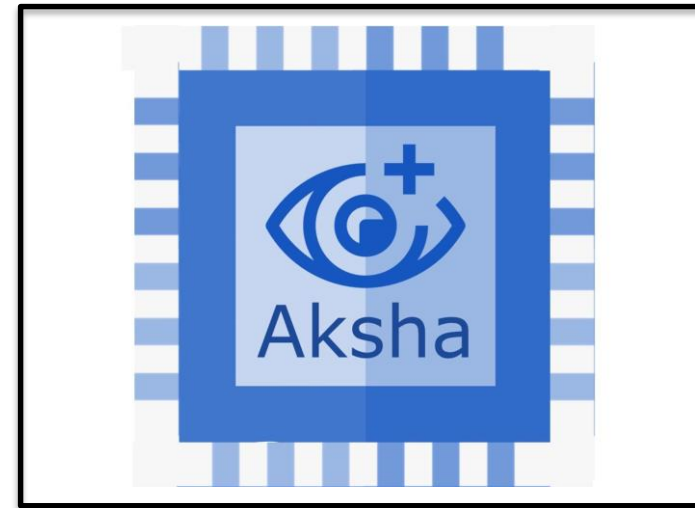
Introduction to Aksha

Overview of Aksha

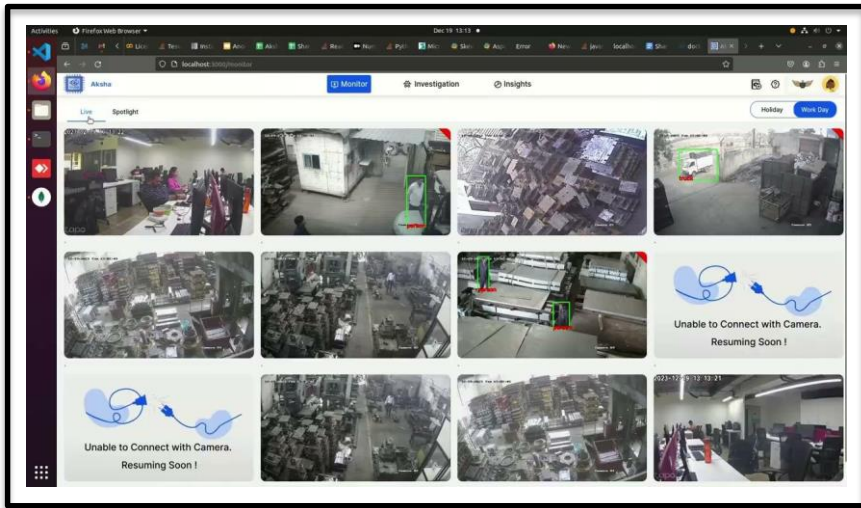
- Aksha is an intelligent surveillance system designed for automated monitoring with minimal human intervention.

- Utilizes advanced technologies like Computer Vision, Artificial Intelligence, and Generative AI for real-time analysis.

- Key functionalities include object detection, anomaly detection, and automated alert generation.



Architectural Approaches



- Offers multiple deployment architectures: On-Premise, Edge, and Cloud solutions.

- Flexibility in UI and backend configurations to suit various operational needs.

- Supports seamless integration with existing infrastructure for enhanced scalability.

Collaboration and Customization

- Collaboration with NDIL led to the development of a customized version focused on worker safety.

- Implemented a three-phase customization process incorporating Generative AI for enhanced features.

- Successful proof of concept demonstrated on Raspberry Pi 4B with dual OS support.

Key Features of Aksha



- Live recording and processing capabilities for real-time surveillance.

- Customizable alerts and insights generated using Generative AI for proactive management.

- Advanced object detection for various entities including vehicles, personnel, and safety gear compliance.

Customer Testimonials

- Clients report significant reductions in human monitoring efforts and enhanced event detection.

- Positive feedback highlights the system's ability to identify abnormal events effectively.

- Users appreciate the cost-effective nature of Aksha in enhancing existing CCTV setups.

Scalability and Performance

- Effortless scalability through microservices architecture using Kubernetes.

- Real-time data processing capabilities ensure timely insights and actions.

- High availability and fault tolerance achieved through decoupled architecture and distributed messaging with Kafka.